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Review article

A systematic review of technologically enhanced assessment of vessel wall movement and mechanical function in dissected and healthy aortas ☆,☆☆

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ABSTRACT

Acute aortic dissection is a vascular emergency with unpredictable progression and outcomes. Standard imaging modalities, such as computed tomography angiography (CTA), facilitate diagnosis and operative planning but provide limited data on vessel wall mechanics and hemodynamics. Modern imaging and computational models may offer deeper insights into natural history prognostication and risk prediction. This PROSPERO-registered systematic review (CRD42024589703) followed the Preferred Reporting Items for Systematic Reviews and Meta-analysis guidelines. Eligible MEDLINE and Embase studies assessed vessel wall dynamics or hemodynamic modeling in healthy or dissected aortas. The risk of bias was assessed using Risk Of Bias In Non-Randomized Studies – of Interventions; the quality of evidence was evaluated with Grading of Recommendations, Assessment, Development and Evaluation. A total of 36 studies were included, mainly retrospective single-center cohort studies of CT/CTA, magnetic resonance imaging, ultrasound, echocardiography, and digital subtraction angiography. An entry tear size greater than 10 mm on CTA was associated with increased false lumen flow and higher sheer wall stress, causing poor aortic remodeling, as confirmed with 4-dimensional magnetic resonance imaging. Distal entry tears and presence of more entry tears reduced the risk of pulsatile-type dissections, having a long-term protec-

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tive effect. Helical flow patterns assessed through computational fluid dynamics determining kinetic energy transmission increase proximal false lumen thrombosis with favorable outcomes. Modern imaging and modeling techniques demonstrate potential in predicting disease progression, with the possibility of tailoring aortic dissection management. Because most studies were exploratory in design, they could not establish replicable clinical protocols or measurable parameters. Measurements of false lumen flow, kinetic energy, and entry tear features (size and number) have demonstrated the greatest potential. Further prospective, standardized studies are essential to translate these findings into routine clinical care.

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1. Introduction

Aortic dissection is the most common aortic emergency, affecting 3 to 10 persons per 100,000 annually [1]. Of these, between 20% and 30% die before hospital admission [2]. It exists on a spectrum of acute aortic syndromes, which comprise type A and type B aortic dissections (TAAD and TBAD), as well as intramural hematomas (IMHs) and penetrating aortic ulcers (PAUs) [2–4].

The natural history of untreated aortic dissections is poorly understood, particularly as it transitions from the acute to the subacute stage and finally to the chronic stage, with relatively high mortality rates. Data from the International Registry of Acute Aortic Dissection revealed that medically managed patients with TAAD had a high in-hospital mortality of 50%, although those with TBAD had a 10% mortality, and of those who survived the initial hospital admission, 88.6% of TAAD and 90.3% of TBAD survived at 1 year [5]. The goals in medical and surgical management are to maintain distal perfusion and promote favorable remodeling to reduce the risk of aneurysmal degeneration and rupture. Optimal medical therapy achieves this with intensive anti-impulse and antihypertension therapy using β-blockade [6,7]. Current models to determine the risk of failure of medical therapy rely on anatomical risk criteria that may impact vessel mechanics in ways that lead to aortic degeneration. Current imaging modalities, however, are not able to adequately identify the dynamic flow forces that occur in dissections and assess how they can lead to aortic wall degeneration.

Although the surgical treatment of TAAD remains open repair, the advent of thoracic endovascular aorta repair (TEVAR) for TBAD was first described by Dake et al in 1999 and has revolutionized management of descending aortic pathology [8–10]. The European Society for Vascular Surgery guidelines recommend TEVAR in the acute setting for complicated TBAD, for chronic TBAD with acute aortic symptoms, and for chronic TEVAR with descending thoracic aortic diameter greater than 60 mm (may be considered between 56 and 59 mm) [6]. Some advocate for the use of TEVAR for uncomplicated TBAD, IMH, and PAU, as several studies have indicated that high-risk radiological and clinical features suggest a potential benefit of early intervention [11,12]. Despite being minimally invasive, TEVAR still has risks [9]. Evidence regarding high-risk features is not robust, despite the increasing technology of diagnostic tools.

Computed tomography angiography (CTA) is the standard of care in diagnosis and operative planning, but it does not provide information on the dynamics of the dissected aorta [6]. With the goal of risk prognostication, a comprehensive assessment of fluid and vessel physics is required, including assessment of kinetic energy transmission in an aorta. The application of imaging technologies that are capable of defining the forces applied to a dissected aorta may be able to inform the timing of a surgical intervention and provide guidance on TEVAR planning and the use of surgical adjuncts that assist in promoting adequate aortic remodeling.

A universal understanding of vessel wall mechanics with dynamic flow changes requires an understanding of aortic changes at various stages, from the pre-dissected aorta to chronic TBAD. Investigation of flow dynamics in a healthy aorta could also assist in informing the impacts of flow disruptions that occur in dissected aortas by acting as a baseline in aortic mechanics. Comparison of vessel mechanics between healthy and diseased aortas may assist in defining the benefits of intervention strategies in returning these mechanics to their baseline after both medical and surgical treatments.

This systematic review aims to summarize the current knowledge of different technologically enhanced assessment methods in patients with dissected and healthy aortas to determine the features of vessel wall movements and mechanical function, answering the following 2 research questions in Population, Intervention, Control, Outcome, Study format:

- In patients with healthy and dissected aortas at any chronicity of disease, what imaging modalities can characterize vessel wall mechanics to define the clinical progression of aortic dissections?
- Are there any physics models derived from imaging which represent the dynamic function of the aortic wall in healthy and dissected aortas at any chronicity of disease to assist in the prognostication of aortic remodeling or deterioration?

2. Methods

A systematic review was performed using protocols defined by Preferred Reporting Items for Systematic Reviews and Meta-

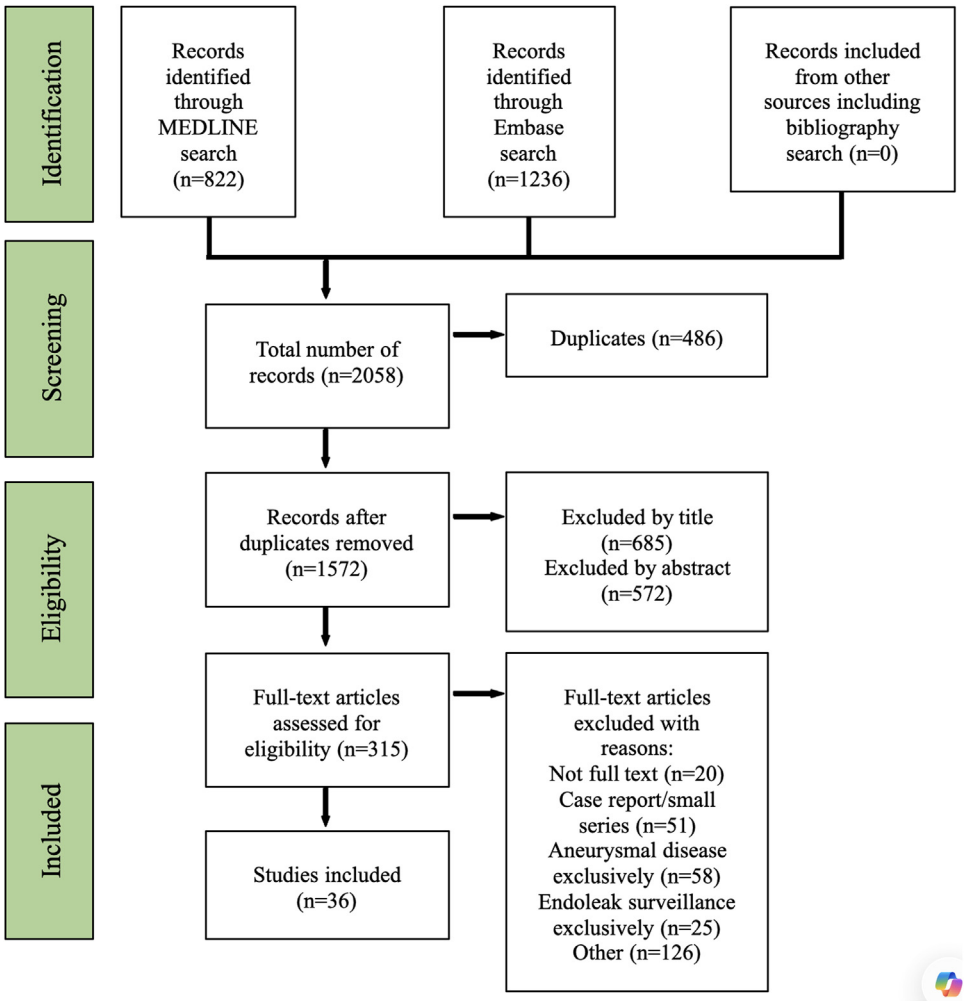


Fig. 1 – Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA-S) diagram summarizing the number of studies produced from the search strategy based on the database queried (Table 1) and application of the inclusion and exclusion criteria (Table 2).

analysis (PRISMA-S) guidelines, summarized in a PRISMA diagram (Fig. 1) [13]. This study's methodology was prospectively registered in PROSPERO with ID number CRD42024589703, and a PRISMA checklist is provided in Supplementary Table 1 [14].

2.1. *Criteria for selected studies*

A search strategy was used (Table 1) with inclusion and exclusion criteria implemented (Table 2). The population arm focused on patients with both healthy and dissected aortas at any chronicity of disease, whereas the intervention focused on imaging modalities to model aortic fluid dynamics. The outcomes focused on vessel wall physics, flow/fluid dynamics, and hemodynamics. MeSH terms were exploded and included in the search.

Exclusion was performed based on article design, interventions, cohort, protocol, and statistics through a title and abstract review. Articles that were only abstracts, non-English, or conference presentations were excluded. Studies that were primarily narrative reviews or systematic reviews were excluded, but their sources were assessed for inclusion. Case reports and series without a comparator were excluded. Studies involving dissections that included any portion of the aorta were included, but those that were exclusively in other arteries were excluded. Studies performed on aberrant anatomy (such as the Kommerell diverticula) were excluded. For each article, a well-defined and repeatable intervention measure (in the form of an imaging modality) was assessed, and it was determined whether statistical significance was calculated using a reproducible and well-described statistical method. Included research had to be published in a peer-reviewed journal and not be redacted.

2.2. *Search method for identification of studies*

A period search of Ovid MEDLINE 1946 and Embase Classic+Embase 1947 using Wolters Kluwer Ovid (Alphen aan den Rijn, the Netherlands) was performed from inception to November 9, 2024. The Cochrane Database of Systematic Reviews was not included because there were no results in a preliminary search. Other technical research databases were

Table 1 – Search strategy based on a Population, Intervention, Control, Outcome, Study design model focusing on population, intervention, and outcome measures. Databases queried were Ovid MEDLINE 1946 and Embase Classic+Embase 1947 using Wolters Kluwer Ovid (Alphen aan den Rijn, Netherlands) with a period search from inception to November 9, 2024.

	Population	Intervention	Outcome	
	1. (aort* adj4 (aneurys* or dissect* or ruptur* or trauma* or tear* or split*)).ti,ab. 2. (Type B adj3 dissect*).ti,ab. 3. (Stanford adj3 dissect*).ti,ab. 4. dissecting aneurysm/	6. diagnostic imaging/ 7. Angiography/ 8. ((magnetic resonance or MR or MRI or NMR) adj5 (angiogra$ or arteriogra$)).ti,ab. 9. ((CT or CAT) adj5 (angiogra$ or arteriogra$ or tomograph$)).ti,ab. 10. (comput$ adj3 tomogra$).ti,ab. 11. ultrasonogra$.ti,ab. 12. (USS or DUS or CDUS or CEUS).ti,ab. 13. radionuclide imaging/	15. ((vessel or arter*) adj3 wall adj5 (physi* or stress or tension or flow)).ti,ab. 16. (fluid dynami*).ti,ab. 17. (blood adj5 flow adj3 mode*).ti,ab. 18. flow dynami*.ti,ab. 19. (aort* adj3 flow).ti,ab. 20. (hemodynami* or haemodynami*).ti,ab.	
	5. OR (1–4)	14. OR (6–13)	21. OR (15–20)	
MEDLINE	67,424	627,547	356,564	822
Embase	92,592	1,064,449	325,674	1,236
			5 AND 14 AND 21	2,058

Abbreviation: OR, odds ratio.

Table 2 – Inclusion and exclusion criteria defined before the search that were applied to the resultant search by 2 independent authors.

	Inclusion	Exclusion
Article	Peer-reviewed publication	Abstract only Non-English article
Design	Randomized controlled trial Prospective and retrospective cohort studies Case series (with comparison papers)	Isolated case reports Case series (no comparison papers) Opinion
Intervention	Imaging modalities	Theoretical papers without patient imaging Invasive interventions
Cohort	Patients with healthy and/or dissected aortas	Patients with aberrant anatomy Patients with dissection through other parts of the arterial tree
Description of protocol	Well-described intervention/treatment measure for all cases	Nonreproducible protocol
Statistics	Valid statistical analysis of results	Inadequately rigorous analysis of results Inappropriate statistical methods

not included because they lack clinical outcomes data, which is a required component of the research questions being investigated.

2.3. *Data collection and analysis*

All results from each database search were imported into Endnote 20 (Clarivate), with duplicates merged. Titles and abstracts were then independently evaluated by authors ECS and CH. Two selection rounds selected the final articles for inclusion, with discrepancies independently evaluated by author WY. Data were extracted by ECS, whereas WY and CH evaluated the final tables for accuracy.

Data extraction was summarized using the Oxford Centre for Evidence-Based Medicine levels of evidence (Table 3) [15]:

- Study type
- Funding
- Population
- Sample size
- Intervention
- Control (if any)
- Outcomes, including statistical analysis if relevant.

A meta-analysis was not performed due to significant heterogeneity in cohorts, interventions, and outcome measures.

The risk of bias was assessed based on the Risk Of Bias In Non-Randomized Studies – of Interventions (ROBINS-I) tool [50–52]. Confidence in cumulative evidence was based on the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach [50,52,53]. Furthermore, the following 7 criteria were used to assess for lower quality evidence: confounding, selection bias, intervention classification bias, performance bias, missing data, measurement bias, and reporting bias.

3. Results

A total of 1,572 studies were identified after duplicates were removed. After title and abstract exclusion, 315 full-text studies were identified for review. Studies were then excluded due

Table 3 – Description of included studies with quantitative data including the Oxford 2011 Levels of Evidence (LOE), categorized based on subgroup.Authors.

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
AI-based algorithms (including computed tomography and magnetic resonance imaging)									
Mayer et al [16]	2024	2b	Retrospective single-center cohort	TU Graz LEAD Project; Austrian Science Fund	All TBAD; patent or partially patent false lumen	40 patients	Manual segmentation of CTA of TBAD	No control arm.	Good intra-examiner correlation
Zhang et al [17]	2023	2b	Retrospective multicenter cohort	National Natural Science Foundation of China; Beijing Natural Science Foundation	TBAD treated with TEVAR with at least 2 follow-up CT	133 patients with 306 postoperative CT	Multi-stage segmentation framework using machine learning to assess postoperative vascular volume quantifications	Previously published dissection segmentation methods	Resolution of thrombus and false lumen not enlarged.
CT including CFD modeling									
Cheng et al [18]	2008	2b	Retrospective single-center cohort	Not specified	Uncomplicated TBAD followed for 12 mo	12 patients	Computational model of pulsatile blood flow through a thoracic stent graft	No control arm	Relationship between drag forces and stent graft area due to a significant change after repair
Cheng et al [19]	2013	2b	Retrospective single-center cohort	Not specified	Acute TBAD	4 acute patients and 1 with follow-up	Pulsatile flow simulations	No control arm	Flow rate in the false lumen was dependent on the size of the entry tear. High shear wall stress at the point of the entry. Progression to false lumen thrombosis associated with prolonged partial residence time.
Hohri et al [20]	2021	3b	Retrospective single-center case-control study	Not stated	Acute TAAD with pre-dissection CTA present	3 patients with AAAD and 3 healthy controls	CFD modeling for OSI leading to acute TAAD	Healthy control patients without cardiac history or hypertension	Vortex flow in the lesser curve of the arch at mid-systole which expands to the entire aorta in diastole. In pre-dissection patients, there was accelerated transvalvular flow at the posterolateral ascending aorta. High shear stress on greater curve in systole. Higher OSI on lesser curve which becomes point of entry tear.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Jiang et al [21]	2023	2b	Retrospective single-center cohort study	Not stated	Complicated TBAD treated with physician-modified endograft	12 models from 6 patients (pre- and post-intervention)	Hemodynamic indicators: TAWSS, RRT, and ECAP	Pre-intervention model	Flow rates to visceral vessels increased after intervention. Average RRT and ECAP decreased. Pressure and TAWSS significantly reduced in postoperative visceral regions.
Li et al [22]	2023	3b	Retrospective single-center case-control study	National Science Foundation of China, Applied Basic Research Program of Sichuan Province, Mao Zi Program of Sichuan Province	Hyperacute TBAD	15 hyperacute TBAD	Flow features and hemodynamic parameters such as wall shear stress and helicity assessed	12 healthy control patients	Mean NtransWSS levels correlated with primary entry tear location. Negative correlation between helicity and NtransWSS. Loss of helicity in the arch and descending aorta may be related to risk of TBAD.
Murayama et al [23]	2010	3b	Prospective single-center case-control study	Not stated.	Double-barrel type TBAD	10 patients	ECG-gated multislice CT	10 healthy controls	Able to differentiate pulsing-type and static-type dissections where pulsing-type is more common if entry tear is proximal without retrograde filling of the false lumen. This can lead to luminal enlargement.
Osswald et al [22]	2017	3b	Retrospective single-center case-control study	None	TBAD with later RTAD	10 patients	CTA to assess pressures, velocity magnitudes, and WSS at the site of the future RTAD entry tear	10 patients with TBAD and no future events	Significantly elevated WSS at the site of the future RTAD entry tear

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Parker et al [24,25]	2022	2b	Retrospective single-center cohort study	Not specified	Acute TBAD	69 patients	False lumen thrombosis morphology based on CTA then assessment of luminal pressure differences	No control arm	Proximal false lumen thrombosis had lower false lumen pressures. Both associated with fewer aortic complications in 1-y follow-up.
Polanczyk et al [26]	2021	2b	Retrospective single-center cohort study	None	Acute TBAD treated operatively	12 male patients	CFD to assess displacement forces on the aortic wall at the location of endograft placement	No control arm	Low velocity and shear rate was associated with higher blood viscosity. Velocity gradient near tear surface increased WSS potentially leading to decreased resistance.
Rinaudo et al [27]	2014	2b	Retrospective single-center cohort study	Fondazione RiMED	TBAD with patent false lumen	25 patients	CFD to assess flow in the false lumen with acute complications and chronic aneurysm formation as outcomes	No control arm	Helical and slow moving flow in the false lumen. Larger entry tear was associated with greater flow and poorer outcomes.
Takeda et al [28]	2022	3b	Retrospective single-center case-control study	Not stated	Aortic dissection (TAAD or TBAD)	8 dissected patients	CFD used to correlate different hemodynamic factors	3 healthy aortas	Flow velocity and WSS correlated strongly. Higher levels in type A dissection at systole.
Weber et al [29]	2009	2b	Prospective database single-center cohort study	Not stated	Chronic TBAD treated with optimal medical therapy.	10 consecutive patients	4D CT using 20 points in an R-R interval with ECG gating	No control arm	More distinct pulsatility of the true lumen. 10%–20% oversizing of endograft likely adequate.
Zhu et al [30]	2021	2b	Retrospective multicenter cohort study	Not stated	TAAD treated surgically	17 patients with or without aneurysmal progression	Morphological and hemodynamic analysis	No control arm	Fewer number of re-entry tears and higher false/true lumen pressure difference related to aneurysmal progression.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
CT renal perfusion									
Liu et al [31]	2017	2b	Prospective single-center cohort study	No funding	Aortic dissection confirmed by TOE	43 patients	Preoperative CT renal perfusion	No control arm.	Blood flow of bilateral renal perfusion was related to the type of dissection, number of entry tears, and false lumen thrombosis TAAD had significantly worse blood flow. No difference in number of intimal tears. Worse flow through false lumen compared with true lumen. Larger entry tears had better flow.
DSA									
Augustin et al [32]	2022	2b	Retrospective single-center cohort	Open Access funding by Projekt DEAL	Aortic dissection and malperfusion syndromes treated with endovascular aortic fenestration	25 patients, 10 from a previous study population	Post-processed color-coded DSA images before and after fenestration with dTTP analyzed	sBP gradient	Significant decrease found in dTTP after fenestration with similar reduction in sBP gradients. Moderate correlation between 2 measures. Mean TTP decreased significantly to 4.96 ± 0.94 s ($P < .001$). Correlation between dTTP and sBP gradients of $r = 0.226$ and $P = .351$.
Guthaner [33]	1983	2b	Prospective single-center cohort	Not stated	Patients with aortic dissection after open repair	6 patients	DSA	Aortography in the form of CTA	Differential flow could be clearly demonstrated on DSA. Can reveal the postoperative fate of the distal false channel.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Echocardiogram –TTE and TOE									
Kang et al [34]	1998	2b	Prospective single-center cohort	Not stated	Patients with suspected aortic dissection	27 patients with IMH and 73 with acute aortic dissections.	TOE	CT	Correlation between two modalities. 3 cases were conflicting: 1 IMH on TOE was atherosclerotic thickening on CT which progressed to dissection; 2 cases of dissection on CT had TEE show crescentic thickening and missed the short segment of intimal flap in a tortuous ascending aorta and abdominal aorta. Sensitivity and specificity of IMH on TOE was 27/27 (100%) and 20/22 (99%).
Roudaut et al [35]	1992	2b	Prospective single-center cohort study	Not stated	Patients who underwent a surgical repair of TAAD	107 patients of which 4 had Marfan's disease	TTE/TOE	CT	Aortic root diameters different between CT and TOE likely due to tangential measurements in CT. Similar to mid-ascending aorta. No difference in upper descending aorta size, intimal tears visible on TOE, and flow. $P < .05$ for aortic root and mid-ascending aorta measurements. Chi-squared test with 2 degrees of freedom used for analysis of variance.
Wiet et al [36]	1994	2b	Retrospective single-center cohort study	Not stated	Patients with suspected thoracic aortic disease or incidentally found in other procedures	199 Patients. 132 for suspected thoracic aortic disease. 67 for incidentally found thoracic aortic disease. 37 Patients referred for suspected aortic dissection.	TOE	CT	CT unable to assess the aortic valve for aortic insufficiency, determine site of interluminal communication or velocity of flow through true and false lumen.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
MRI and MRA									
Chen et al [37]	2021	2b	Prospective single-center cohort study	Chang-Gung Memorial Hospital	4D MRI of aortic dissections	32 patients with any dissection; 6 patients with TBAD	Hemodynamic parameters of true and false lumen	No control arm	Higher stroke volume in true lumen for patients with stent grafts than those without. Regurgitant flow (determining nonlaminar flow) higher in the false lumen.
Cosset et al [38]	2022	2b	Prospective single-center cohort study	Not stated	4D MRI pre- and post-operatively for TBAD	7 patients with TBAD undergoing elective TEVAR	Helical flow in the false lumen, reverse flows and stasis at various locations	No control arm	Proximal helical flow in the false lumen disappeared after TEVAR but appeared distally in some cases, particularly at secondary entry tears. Forward flow increased after TEVAR in the true lumen.
Engel et al [39]	2024	2b	Retrospective and prospective single-center cohort study	American Heart Association	4D MRI of TBAD measured 120 d apart	32 patients, 22 retrospectively, 10 prospectively	True lumen and false lumen total kinetic energy, maximum velocity, mean forward flow, and mean reverse flow	No control arm	Baseline aortic diameter did not correlate with growth rate. False lumen maximum velocity and true lumen kinetic energy did correlate with growth rate.
Kinner et al [40]	2015	2b	Prospective single-center cohort study	Not stated	Contrast-enhanced 4D MRA and 3D MRA for TBAD	19 patients	Image quality and effect on decision-making	No control arm	10/19 patients with flow assessment of visceral vessels had management decisions changed.
Kunz et al. [41]	2004	2b	Prospective single-center longitudinal cohort study	Not stated	Contrast-enhanced 3D MRA ν 3D MRI in aortic dissection	53 patients (12 TBAD)	Supra-aortic branch vessel involvement and intimal flap identification	No control arm	Contrast-enhanced MRA significantly superior in assessment.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Liu et al [42]	2018	2b	Prospective single-center cohort study	Not stated	4D MRI for TBAD	16 patients	Hemodynamic parameters in true and false lumen	CTA	Intimal flap entry size negatively correlated with true lumen flow velocity and flow rate and positive correlation with average through-plane velocity, average net flow, and peak flow in the false lumen. Blood flow indices enhanced in true lumen with multiple entry tears and peak flow lower in false lumen. In false lumen thrombosis, increased average through-plane velocity and peak velocity magnitude in the true lumen.
Strotzer et al [43]	2000	2b	Prospective single-center cohort study	Not stated	3D MRA in aortic dissection	14 patients	MRA performed with the 3D FISP sequence	No control arm	Significant variation in flow properties in the dissected aorta. There is rarely bidirectional flow in the true lumen but common in the false lumen with wide range of reflux.
Trojan et al [44]	2017	2b	Prospective single-center longitudinal cohort study	Deutsche Forschungs-gemeinschaft and Ruprecht-Karls-Universit at Heidelberg	3D MRA for chronic TBAD	20 patients with chronic or residual TBAD and patent false lumen	Hemodynamic properties comparing stable and expansion cohorts	No control arm	Time-to-peak intensity and full width at half maximum were significantly shorter in the expansion group. Upward and downward slopes in time-intensity curves steeper in the expansion cohort.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Multi-modal analysis/review									
Nienaber et al [45]	1993	2b	Prospective single-center cohort study	Not stated	Suspected dissection of the thoracic aorta	110 patients. 47 had all tests, 97 had 3 or more. All had at least.2 procedures	TTE then TOE, CT, or MRI	Comparison between modalities	TOE had best sensitivity with 100% scores for acute/subacute type B dissection. CT and MRI similar results but higher for MRI in type A. TTE sensitivity low for type A acute at 78.3% but lower in type B acute at 40%. TEE had highest sensitivity for both type A and type B for diagnosis. TTE specificity excellent compared with others.
Ruiz-Muñoz et al [46]	2024	2b	Prospective single-center cohort study	Instituto de Salud Carlos III; Spanish Ministry of Science; Innovation and Universities; Spanish Society of Cardiology; Biomedical Research Networking Centre on Cardiovascular Diseases	Patients with chronic non-thrombosed or partially thrombosed false lumens in the descending aorta after aortic dissection	68 patients included	CTA, MRA and 4D MRI	No control arm	False lumen kinetic energy is related to the presence and extent of false lumen thrombosis. Segmentation performed by MRA and entry tear identified on CTA.
Tomasi et al [47]	2020	3b	Case series with comparison	In partnership with Ansys Inc. without funding	Uncomplicated chronic type B dissection and healthy aorta	2 patients compared with each other	0D-3D coupled model with CTA and MRI	No control arm	Blood pressure of the false lumen less important than true lumen. Aortic wall shear stress and velocity fields increase at the entry tears. Correlates with changes in sBP.

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Table 3 (*continued*)

	Year	LOE	Study type	Funding	Population	Sample size	Intervention	Control	Outcomes/Statistics
Wang et al [48]	2024	3b	Case series with comparison	National Natural Science Foundation of China; Science and Technology Commission of Shanghai Municipality; Shanghai Municipal Health Commission's Health Youth Talent Training Program; Clinical Medicine Innovation Fund of Fudan Zhangjiang Institute; Royal Society, UK; Imperial College London BHF Centre of Research Excellence	Type B aortic dissection receiving TEVAR	5 patients	ECG + SV or just stroke volume. All combined with computational fluid dynamics by gaining boundary inlet conditions.	CFD combined with 4D MRI for boundary inlet conditions.	ECG + SV successfully showed flow distribution in the descending aorta and also predicted wall shear stress in the primary entry tear.
Ultrasound									
Clevert et al [49]	2008	2b	Prospective single-center cohort study	Not stated	Patients with suspected treated and untreated aortic lesions detected by CTA	36 patients. 9 cases were aortic dissection	CCDS and CEUS	Multi-slice CTA	CEUS clearly revealed the true lumen in all cases. CCDS only in 6 cases and in 3 could not detect the false lumen. Renal perfusion could be monitored with CEUS. Significant difference between CCDS and CEUS with McNemar's test $P < .0001$

Abbreviations: AI, artificial intelligence; CCDS, color-coded duplex sonography; CEUS, contrast-enhanced ultrasound; CFD, computational fluid dynamic; CT, computed tomography; CTA, computed tomography angiography; DSA, digital subtraction angiography; dTTP, difference in time to peak; ECAP, endothelial cell activation potential; ECG, electrocardiogram; MRA, magnetic resonance angiography; MRI, magnetic resonance imaging; NtransWSS, normal transverse wall shear stress; OSI, oscillatory shear index; RRT, relative residence time; RTAD, retrograde type A dissection; sBP, systolic blood pressure; SV, stroke volume; TAAD, type A aortic dissection; TAWSS, time-averaged wall shear stress; TBAD, type B aortic dissection; TEVAR, thoracic endovascular aortic repair; TOE, transesophageal echocardiogram; TTE, transthoracic echocardiogram.

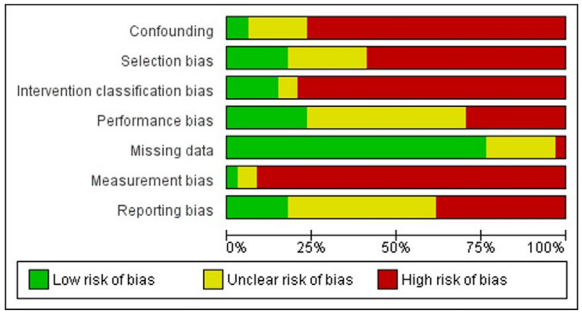


Fig. 2 – Risk of bias graph of each risk of bias item presented as percentages across all included studies based on the Risk Of Bias In Non-Randomized Studies – of Interventions (ROBINS-I) tool.

to the exclusion criteria, being exclusively aortic aneurysm studies, endoleak-specific studies, or having other issues such as being purely engineering studies. A total of 36 studies were included in the final systematic review.

Most of the included studies were retrospective, single-center, cross-sectional cohort studies and limited longitudinal studies. There were no randomized controlled trials.

A narrative synthesis approach to analysis was performed, using guidelines outlined by the Synthesis Without Meta-analysis framework [55]. This process grouped studies for synthesis, and then the measurement metrics used were reviewed. Synthesis was performed by including a standardized metric from each study that permitted cross-comparison. By isolating population, intervention, and outcomes, including statistical outcomes, a qualitative appraisal of heterogeneity was performed. This allowed for transparent and comprehensive interpretation of the available evidence, despite the inability to perform a statistical quantification of heterogeneity.

There was a significant risk of bias in most studies due to selection and measurement biases; however, methodologies were well described and repeatable in most cohort studies (Figs. 2 and 3). Selection bias was significant in most studies due to the lack of predefined selection criteria for patients within the observational series. This affected the external validity and generalizability of these studies. Measurement bias was introduced due to a lack of standardized measurement metrics across the studies, without the use of a core laboratory to perform these measurements. This limited interstudy comparability; therefore, shared measurements should be interpreted with caution.

3.1. *Imaging modalities for vessel wall mechanics*

All studies included at least 1 imaging modality relevant to the vessel wall mechanics of the aorta. Of these, CT/CTA was the most frequently investigated modality in 13 studies and magnetic resonance imaging (MRI)/magnetic resonance angiography (MRA) in 8 studies. CT renal perfusion was investigated in 1 study, and digital subtraction angiography (DSA) was investigated in 2 studies. Ultrasound was analyzed in 4 studies, including standard ultrasound and echocardiogram.

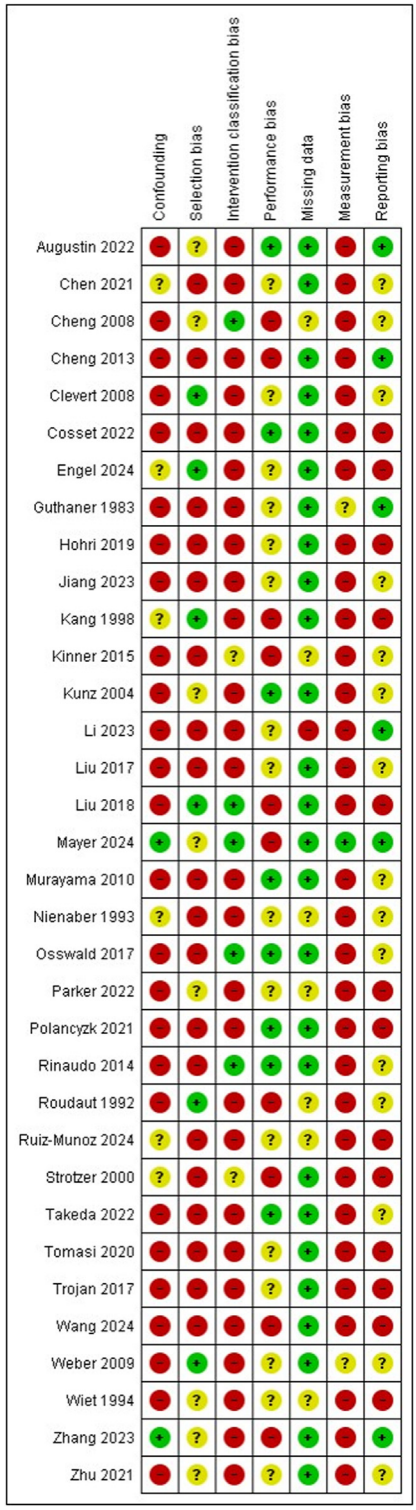


Fig. 3 – Risk of bias summary for each risk of bias item for each included study based on the Risk Of Bias In Non-Randomized Studies – of Interventions (ROBINS-I) tool.

Electrocardiographic (ECG)-gated CTA could reduce the influence of motion artifact due to cardiac pulsation and more reliably describe the impact of vessel wall movements on fluid mechanics. In a study of 10 patients including ECG-gated CTA of the thoracic aorta at 10 different stages of the cardiac cy-

cle, dissections were classified as pulse-type dissections (defined as more than a 125% increase in the maximum area of the true lumen compared with its minimum area) versus static-type dissections, which had less dissection flap movement [23]. The proximal entry tear had a distinct effect on this phenomenon and on energy transmission in pulse-type dissections to the weakened aortic wall, where there was no retrograde flow (compared with 40% in static-type dissections). This was speculated to lead to future false luminal enlargement and to be assessed in future research [23].

In a prospective single-center cohort study assessing specialized MRI protocols in 19 patients, contrast-enhanced 3D MRA, which uses gadolinium-based contrast agents to visualize vascular structures with high spatial resolution, was superior to standard CTA in the assessment of supra-aortic branch vessel involvement and intimal flap identification [40,41]. Contrast-enhanced 4D MRA also uses contrast to add a temporal factor to the 3D MRA to demonstrate flow patterns within vasculature over a period of time. It is capable of describing visceral branch flow after TBAD, with improved flow rates after interventions. This occurs due to a decrease in endothelial cell activation potential, defined as the ratio of oscillatory shear index to time-averaged wall shear stress (WSS) [21,20]. Contrast-enhanced 4D MRA led to changes in aortic management decisions in 10 of 19 patients, where time-based assessment of luminal fluid flows demonstrated delayed visceral organ perfusion missed by CTA [40].

Ultrasound, including contrast-enhanced ultrasound (CEUS) and transthoracic or transesophageal echocardiogram (TTE/TOE), offers real-time evaluation of vessel dynamics. A comparative observational study of 36 patients (9 dissections) comparing CEUS and CTA found that CEUS could identify the pulsating septum, re-entry tear, and true lumen, all verified with CTA [49]. CEUS could also assess dynamic renal perfusion. There were challenges assessing renal perfusion with CTA, whereas color Doppler ultrasound provided early signs of acute kidney injury, resulting in surgical intervention [49,31].

TTE/TOE is user dependent and adds complexity by requiring specialized resources and often anesthetic support in the case of TOE. Most research on the role of TTE/TOE is historical, with no research in the 21st century specific to the research questions. In a cohort of 73 patients with acute aortic dissections, TOE could assess flow in the false lumen and determine no-flow, low-flow, or high-flow dissections and differentiate IMH with a no-flow false lumen (sensitivity and specificity of 100% and 99%, respectively) [34,35,45]. TOE can also provide additional information on flow-related sequelae of aortic dissections, including aortic valve insufficiency, assisting in prognostication [36].

Adjunctive assessments using CT renal perfusion can assess the role of dynamic or reduced renal perfusion caused by a mobile intimal flap or a compressed true lumen. One prospective cohort study comprising 43 patients investigated preoperative CT renal perfusion. Worse flow was found in renal arteries branching off the false lumen, with significantly better flow associated with larger entry tears [31].

Despite being invasive, DSA is effective in evaluating multiple angles of the aorta and providing time-based assessments. Standardized protocols to assess fluid dynamics have not been described. Typically, DSA is reserved for peri-interventional assessment, with an operative plan established. Perioperative imaging with DSA can provide insights into the extent of coverage required and adjunctive interventions. Differential flow and the fate of the distal false lumen could be assessed with DSA in a small cohort of 6 prospectively collected patients [33].

3.2. *Physics models for prognostication*

A complex relationship between fluid dynamics and vessel mechanics correlates with risk prognostication, as demonstrated in several small-scale studies. With a protocolized spiral CT technique, a study comprising 10 patients revealed that computational fluid dynamic (CFD) models can assess kinetic energy transmission to the aortic wall based on false lumen and entry tear morphology. In this study, the difference between static and pulsing type dissections was revealed, with the latter having poorer aortic remodeling [23]. The relationship is strengthened by several CT and MRI studies demonstrating that false lumen flow rate is linked to the size and number of entry tears, leading to greater energy transmission to the aortic wall and poorer outcomes [19,27,42,47]. In a study of 16 patients using 4D MRI which measures particle accelerations to provide temporal flow characteristics of any tissue, the size of the entry tear negatively correlated with peak velocity, average net flow, and peak flow in the true lumen and positively correlated with average net flow and peak flow in the false lumen [42]. Additional studies have revealed that entry tears of less than 10 mm have a false lumen flow rate of 31.6% ($n = 8$) of total flow, whereas those greater than 10 mm had flow rates of 64.5% in the false lumen ($n = 17$) [27]. Multiple entry tears have protective effects, with reduced false/true lumen pressure differentials assessed using CFD based on CTA and MRA in cohorts of 17 and 16 patients, respectively, leading to less aneurysm progression [30,42]. True lumen kinetic energy and maximum flow velocity correlated with growth rate based on 4D MRI within a longitudinal cohort study of 22 retrospectively and 10 prospectively collected patients [39]. Time-to-peak (TTP) intensity from time-resolved 3D contrast-enhanced MRA related to an expansion cohort in TBAD with a patent false lumen in a longitudinal prospective single-center cohort study of 20 patients [44]. Time-intensity curve slopes were steeper in the expansion cohort. TTP was further evaluated with complex post-processed color-coded DSA analysis before and after fenestration of the TBAD septum in a retrospective sample of 25 patients [32]. A significant decrease was found in TTP after fenestration, also reducing systolic blood pressure gradients between true and false lumens.

Several studies have investigated the role of partial false lumen thrombosis on prognostic measurements, with lower false lumen pressure in partial false lumen thrombosis previously described as a high risk factor [24,56]. Proximal false lumen thrombosis had fewer aortic complications in follow-up compared with a patent false lumen, with an odds ratio of 0.17 (95% CI, 0.04 to 0.60) and fewer 1-year adverse aortic events, with a hazard ratio of 0.36 (95% CI, 0.16 to 0.80) in a study of 69 patients [24,25]. The probability of progressing to thrombosis is related to prolonged partial residence time from a CFD model which describes the length of time a particle remains within a specific measured region before moving. This reflects

a false lumen inflow ratio; values of 24% kinetic energy transference and 0.4 mJ are more likely to develop partial thrombosis in a prospective cohort of 68 patients [19,46]. Furthermore, 4D MRI assessment revealed that in false lumen thrombosis, increased average through-plane velocity and peak velocity magnitude in the true lumen led to favorable outcomes [42]. The placement of an endograft resulted in higher regurgitant flow in the false lumen, leading to nonlaminar flow and subsequent false lumen thrombosis. Significant reflux in the false lumen was detected through 4D MRI in two separate prospective cohort studies comprising 32 and 14 patients [37,43].

In several studies, the relationship of entry tear to flow was critical. Dynamic pressure gradients found in pulse-type dissections, when compared with static dissections, were relevant to the false lumen enlargement and were more likely to require intervention [23]. This was found in dissections with distal entry tears and retrograde dissection not leading to pulsating-type dissections, likely due to reduced pressurization based on entry tear arrangement, in a prospective case-control study of 10 patients with double-barrel type TBAD compared with 10 healthy controls [23]. Pulsatile flow could be assessed using CFD models when correlated with a unique 4D CT protocol able to measure 20 points in an R-R interval using ECG gating within a prospective cohort of 10 consecutive patients. This could assess the pulsatility of the true lumen, with 10% to 20% endograft oversizing found to compensate for this pulsatility [29]. Proximal entry tear also related to the loss of helical flow across the aortic arch, promoting poor aortic remodeling in a cohort of 15 hyperacute TBAD patients compared with 12 healthy controls [22]. Slow-moving false lumen helical flow was associated with entry tears of >10 mm; these had increased false lumen flow rates and poorer outcomes in a single retrospective cohort study of 25 patients with patent false lumen [27].

Several flow models have been found across multiple studies to have prognostic qualities:

- False/true lumen flow rate [19,27,39,42]
- False/true lumen pressure differentials [30,42]
- Slow retrograde flow [18,23,43]
- Time-to-peak intensity [44]
- Time-intensity curve slope [44]
- Relative residence time (RRT) [21,27]
- Pulsatile flow [23,29]
- Helical flow [27,57]
- Turbulent kinetic energy (TKE) [39]

The flow of any viscous Newtonian fluid exhibits drag forces on the aortic/endograft wall which can be assessed using CFD models. A relationship between drag forces and the area in which a stent graft is placed was demonstrated in 12 patients followed for 12 months [18]. Differential flow through different portions of the aorta results in different wall stresses, with potentially different prognostic benefits for aortic complications or negative aortic remodeling. An 11% increase in graft inlet area and a 58% increase in the distal outlet area at 12 months were found. This increased the drag forces of larger grafts by 19.5% with graft expansion [18].

High WSS correlated with flow velocity, where a 12-patient retrospective cohort study revealed an association with higher blood viscosity and thrombosis potential [26]. Comparison of 8 patients with aortic dissection to 3 healthy aortas revealed that flow velocity and WSS correlated with the size of proximal entry tears, potentially leading to aortic dilatation [28]. A negative relationship between helical flow and WSS supports evidence that loss of helical flow through the arch and slowed false lumen helical flow may result in poorer aortic outcomes [22]. When deploying an endograft, a comparison of 10 patients with retrograde type A dissections (RTAD) to 10 patients with TBAD and no future events found that there was significantly elevated WSS at the site of future RTAD [54].

4. Discussion

Risk prognostication in aortic dissections is challenging due to a limited number of high-quality studies providing robust evidence. Advancements in imaging modalities allow for a more refined assessment of hemodynamic parameters, which may aid in prognostication. This systematic review highlights imaging-based prognostic factors in aortic dissection, focusing on hemodynamic features contributing to disease progression and adverse outcomes. Owing to the fundamentally exploratory nature of the studies, quantitative thresholds for prognostication were not described for clinical application but provide a basis for future research to this effect. Although comparability of metrics across studies was limited due to selection and measurement bias, findings were consistent across multiple studies for the impacts of pulsating-type dissections, false lumen thrombosis, and the overall impact of flow patterns on WSS (Table 4).

Dissection characteristics have been identified as prognostically significant. Pulsing-type dissections are associated with larger pressure gradients, predisposing to aortic enlargement [23]. This is closely linked to false lumen filling through proximal entry tears, with effectiveness evaluated through well-protocolized spiral CT. Intraoperatively, using intravascular ultrasound (IVUS) has not been independently investigated for its ability to provide dynamic imaging, particularly for determining entry tears, dynamically changing true and false lumens and distal perfusion, but it is recommended for use in the current aortic arch pathologies expert consensus between the European Association for Cardio-Thoracic Surgery and the European Society for Vascular Surgery [58,59]. Research has not been performed on the use of IVUS for investigating dynamic vessel mechanics due to the invasive nature of the investigation and because it is usually applied in a setting where surgical intervention is planned within the same procedure. Therefore, IVUS has a limited ability to investigate the longitudinal natural history of aortic dissections. MRI and MRA may provide insight into flow patterns, assisting in predicting false lumen thrombosis [37,43]. Reliably measurable regurgitant flow patterns will need to be described and included in MRA protocols before they are included in standard clinical practice. MRI also provides a superior assessment of genetically triggered aortopathies, such as Marfan's syndrome, particularly in complex dilatation, where fluid dynamics may assist in predicting future degeneration [60].

Partial false lumen thrombosis has location-dependent prognostic implications. Proximal false lumen thrombosis

Table 4 – Summary of all measurable prognostic parameters with supporting evidence summarized by factor and imaging modality.

Prognostic factor	Summary	Imaging modality	Sources
Entry tear size and number	False lumen flow rates are dependent on the size of the entry tear with high shear wall stress at this point. Regions of high oscillatory shear index are where the entry tear occurs. Greater number of re-entry tears has less aneurysmal progression.	Computed tomography with computational fluid dynamic modeling. 4D magnetic resonance imaging can also be used to determine the impact of entry tear flow patterns.	Cheng et al [19], Hohri et al [20], Li et al [22], Osswald et al [54], Zhu et al [30], Liu et al [31]
Regurgitant false lumen flow patterns	The location of the entry tear can lead to proximal retrograde filling that does not thrombose the false lumen leading to luminal enlargement.	Electrocardiogram-gated multislice computed tomography.	Murayama et al [23]
Helical flow dynamics	Helical and slow moving flow in the false lumen is associated with smaller entry tears and better outcomes. This flow resolves over TEVAR.	Computed tomography with computational fluid dynamic modeling for flow pattern assessment. This can also be assessed with 4D magnetic resonance imaging.	Rinaudo et al [27], Cosset et al [38]
Kinetic energy distribution	The true lumen kinetic energy correlated with growth rate, particularly when compared with the energy within the false lumen. Within the false lumen, also correlated with false lumen thrombosis.	Computed tomography and magnetic resonance imaging including angiography.	Engel et al [39], Ruiz-Muñoz et al [46]
Partial false lumen thrombosis	Prolonged partial residence time can be derived from entry tear location affecting remodeling.	Computed tomography with computational fluid dynamic modeling.	Cheng et al [19]

Abbreviation: TEVAR, thoracic endovascular aorta repair.

confers a protective effect, whereas partial thrombosis elsewhere is associated with an increased risk of adverse outcomes, with MRA-based hemodynamic modeling assisting in prediction [24,56]. Kinetic energy transfer plays a pivotal role, where CFD modeling, validated through MRA, helps predict the likelihood of thrombosis by analyzing the partial residence times of blood flow. To date, the quantitative relationship between kinetic energy and partial residence time has not been validated [19,39].

TEVAR has become the surgical standard, but the prediction of long-term outcomes remains challenging. Several hemodynamic factors that can be assessed with CTA and MRA contribute to the risk of complications after intervention [61]. Although providing a rigid repair, a stent graft alters drag forces, potentially leading to stent migration and endoleak, factors that can be assessed with MRA [18,37]. The velocity gradient at stent placement also results in WSS alterations, which may increase RTAD risk [54]. Preoperative CFD modeling may provide risk profiling to optimize stent placement and minimize complications, but it has not been confirmed in any clinical studies [26,28].

Anatomical regions prone to high WSS have been identified using CTA-based CFD as critical sites for disease progression. The lesser curve of the aortic arch is susceptible to vortex flow, placing significant mechanical stress on this region, potentially leading to dissection at this site [20,48]. The greater curve of the arch during systole exhibits a higher OSI which can lead to dissection depending on aortic arch shape and configuration [20]. Although only preliminary, the introduction of higher computer processing capacities within radiology services may incidentally identify high-risk patients with a CTA for any indication.

Flow dynamics within the true and false lumens can predict dissection progression, where lower velocity flow within the false lumen is associated with an increased risk of degeneration due to the transfer of kinetic energy to the aortic wall [39]. CEUS is capable of determining blood flow and kinetic energy–transmitting eddy currents, particularly in spiral dissection true lumens [62]. The size and number of entry tears influence flow patterns; larger entry tears facilitate higher flow rates, whereas multiple entry tears have a protective effect [19,27,42].

Helical flow can be modeled through CFD [22]. The presence of true lumen helical flow reduces WSS which may predict improved long-term aortic remodeling and outcomes [22]. With slow false lumen helical flow, particularly with large entry tears, the risk of adverse outcomes increases [27]. Endovascular stenting may inadvertently shift this high-risk flow pattern distally, necessitating careful procedural planning to mitigate unintended consequences [38].

Regions experiencing high WSS and slow flow are more likely to thrombose, leading to endothelial activation, which

results in suboptimal aortic remodeling [21]. These factors indicate the value of incorporating advanced flow assessment techniques, such as MRA and CFD modeling, into clinical decision-making to refine risk stratification and optimize individualized treatment strategies.

CFD integration with advanced imaging provides a detailed analysis of complex relationships between fluid dynamics, entry tear characteristics, and aortic wall stress. Parameters derived from CFD, such as false/true lumen flow rates, pressure gradients, risks of thrombosis, and kinetic energy transmission, are potentially valuable prognostic indicators. Emerging techniques such as 4D MRI and phase-contrast velocimetry may further refine CFD models, allowing more accurate predictions of aneurysmal degeneration based on flow dynamics, helical flow patterns, and turbulent kinetic energy, by developing complex definitions of aortic wall mechanics. Before these can develop clinical applicability, they require validated threshold and cutoff values. Other emerging technologies play a role in aortic dissection evaluation and prognostication, including radionuclide imaging; however, there are no studies investigating their use in determining vessel wall function or fluid dynamics, representing a novel branch of research.

As technology and medicine transition to the age of artificial intelligence (AI) through machine/deep learning, diagnosis and prognostication will attempt to use this poorly understood and implemented technology. Unfortunately, medicine remains behind other industries in adopting novel technologies, particularly in the field of surgery, with only two key studies assessing flow using AI-based algorithms focusing on CT to date. These studies only aimed to determine the intra-examiner reliability, which appears to be adequate, the quantification of vascular volumes within the false lumen, and the formation of thrombus [16,17]. This technology may assist in rapidly determining key prognostic outcomes, but its utility has not yet been demonstrated.

4.1. *Limitations*

This systematic review has several limitations. Most included studies were single-center cohort studies, often retrospective in nature, limiting the generalizability of the findings. There were relatively few longitudinal studies that could provide more robust data on disease progression over time. With most studies investigating a population cross-section or having short follow-up periods, there can only be a limited interpretation of remodeling outcomes. Defining long-term prognostication is challenging with the currently available data. Assessment of the true clinical benefit of advanced imaging in patients with dissection requires further studies with long-term clinical endpoints available. In addition, much of the data from the studies were exploratory in nature and did not provide clinically applicable conclusions, such as measurable metrics through the imaging modalities studied. The studies included did not use standardized and defined measurement metrics which is a major limitation related to interstudy comparability.

In addition, publication bias may be at play, as negative or inconclusive studies may lack interest or impact for publication. Heterogeneity among the included studies in terms of imaging protocols, patient selection criteria, and follow-up durations may also affect the comparability of the findings.

5. Conclusions

This systematic review highlights the advancements in imaging modalities for assessing vessel wall movement and hemodynamics in aortic dissection in the context of risk prognostication in this dangerous and poorly understood pathology. Technologies such as CT, MRA, and CFD can provide measurements of flow dynamics, WSS, and/or physical properties of flow and vessel wall mechanics that influence aortic remodeling and have prognostic benefit. The clinical utility of these measurements has not been evaluated. Although most included studies were of a low level of evidence, several measurable imaging parameters have demonstrated prognostic potential. These include entry tear size and number, regurgitant false lumen flow patterns, and helical flow dynamics derived from false lumen flow and kinetic energy, which may serve as early markers of adverse disease progression. Furthermore, partial false lumen thrombosis and its location can be predicted based on entry tear characteristics and kinetic energy distribution. Further focused research and higher-quality evidence are needed before these imaging-based assessments can be integrated into clinical decision-making. Standardized imaging protocols and advanced computational modeling using CTA and MRA may support individualized treatment strategies, enhancing prognostication and long-term outcomes in patients with aortic dissection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1053/j.semvascsurg.2026.06.003.

CRediT authorship contribution statement

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